Which Listener? Audience Matching for Pragmatic Legibility Rewards in Multi-Agent Reinforcement Learning

Anonymous Author(s)

ABSTRACT

Cooperative reinforcement learning agents often fail because their intentions are illegible to their partners, not because they act poorly. We study an intrinsic reward for legibility, imported from the Rational Speech Act (RSA) model of pragmatics through its structural correspondence with maximum-entropy RL. The reward is a listener's log-posterior over an agent's private state given its behavior, computed offline with no execution-time cost, and its decisive design choice is which literal listener L_0 anchors it. We show that any listener makes the reward a variational bound on the intent–behavior mutual information, and that a listener which decodes confidently supplies little shaping gradient. On *scout-support*, a benchmark whose oracle-gap criterion certifies that communication has task value, hand-crafted motion-grounded listeners close 61–82% of the oracle premium (Welch $t \geq 4$; the 82% endpoint at t=3.98 survives Bonferroni). Learned listeners instead fail by succeeding: trained to convergence they decode any behavior, saturate, and shape nothing, and bounding them to the partner's viewpoint does not help. RSA recursion recovers part of the loss (27%, held-out confirmation). The resolution is the *inverse-planning listener*, which Bayes-inverts a frozen non-communicative baseline policy; one RSA step then closes 66–70% ($t \geq 5.8$), matching the best hand-crafted listener with no task-specific geometry. An *audience-matching principle* emerges: anchor the reward in a listener no stronger than the intended reader and crediting only signals that reader can use. The inverse-planning listener satisfies it by construction. We map where the effect holds (gain $\approx$ 0.5× premium) and extend the reward to continuous meaning spaces and to an agent's own bottlenecked decision latent, whose behavioral readability the reward raises without altering its content (suggestive tier).

CCS CONCEPTS

• **Computing methodologies → Multi-agent systems**.

KEYWORDS

multi-agent reinforcement learning; legibility; intrinsic motivation; computational pragmatics; implicit communication

1 INTRODUCTION

In cooperative multi-agent reinforcement learning (MARL), an agent's success depends not only on its own actions but on being *understood* by its teammates: an agent that rushes along an ambiguous path can cause a partner to abandon a target it never intended to claim, while a slightly suboptimal path that unambiguously singles out the intended target enables better coordination. This property is called legibility [1], and standard RL agents, optimized solely for task reward, have no incentive to possess it.

This work studies an intrinsic reward R_{comm} for communicative clarity. We model an observer that reasons pragmatically about *why* the agent chose its action over the alternatives, using the Rational Speech Act (RSA) framework [2]. Any such observer rests on a *literal listener* L_0, the common ground specifying how behavior is interpreted absent pragmatic reasoning. Our central claim is that this choice is not an implementation detail but the decisive factor in whether legibility rewards help or hurt, and that it fails at both extremes. A hand-crafted listener can be misaligned with what the policy can express, incentivizing costly signaling no observer decodes. A listener learned from the agents' own behavior guarantees decodability, but trained to convergence it decodes *everything* and demands nothing: its reward saturates, and a gradient-saturation bound formalizes the failure. Mapping the productive region between these extremes is the empirical core of this paper. Our contributions:

1. **A pragmatic derivation with theory locating the listener space.** A structural correspondence between Max-Ent RL and RSA—both softmax choice models—imports the RSA listener value as an offline intrinsic reward (CTDE-compatible; purely communicative models such as CRSA [3] occupy the myopic corner of the same objective). *Any* listener makes the expected reward a variational lower bound on $I(M;S,A)$ (ML training merely tightens it) [4, 13]; a *gradient-saturation* bound shows a listener decoding with high probability shapes correspondingly little—the tightest bound can be the worst reward; a *vocabulary-restriction* proposition shows a listener factoring through reader-computable statistics can only pay for codes the reader can extract. Together these give the audience-matching principle's two clauses their formal content: (a) non-saturation under the training distribution, (b) factoring through reader-visible statistics.
2. **An evaluation protocol for communicative rewards.** The *oracle-gap* criterion gates benchmarks on whether communication has task value at all; *scout-support* makes task-optimal motion uninformative exactly when information is needed; a blind-partner control separates communication from shaping; and a documented pre-convergence hazard shows why converged budgets are non-negotiable for co-adaptive measures.
3. **An empirical map with an audience-matching principle and a constructive recipe.** Hand-crafted motion-grounded listeners close 61–82% of the premium (Welch $t \geq 4$, frozen hyperparameters, three variants); co-adapting learned listeners saturate and shape nothing—a failure surviving input and window bounds, RSA recursion restoring a quarter; the *inverse-planning listener*—the classical Boltzmann observer self-instantiated from a frozen non-communicative baseline—closes 66–70% with

one RSA step ($t \geq 5.8$), matching the best hand-crafted listener with no task-specific geometry and satisfying both principle clauses by construction. Premium-vs-cost sweeps yield a proportionality regularity with characterized failure modes; on a continuous meaning space the literal listener stays inert and the recursion's advantage never reverses sign while premiums shrink—the meaning-space axis of the map.

(4) **Legible latents: meanings discovered rather than specified.** Routing private observation through a variational bottleneck and aiming the particle recursion at the agent's *own* decision latent yields, to our knowledge, the first oracle-certified premium for exposing a *learned* latent whose content the designer never specified—task pressure selects it, decoy-verified—while the legibility reward raises the latent's behavioral readability (transparency .340→.351) leaving content and information budget untouched. Closures in this regime are suggestive-tier under our own policy; the certified elements are the premium and the measurement protocol.

2 RELATED WORK

Legibility and observer-aware rewards. Legibility inverts goal inference [14, 15, 51]: the actor shapes behavior so an observer's inference succeeds. Formalized for robot motion [1], it extends to sequential tasks (including goal-posterior rewards from Boltzmann-inverted per-goal optimal values [16], the construction our inverse-planning listener ports to deep MARL), observer-aware MDPs [17, 18], and multiagent RL via belief-gain shaping [19]. Observer models as in-RL reward signals have precedent: policy search against human observers [38], observer-in-the-loop shaping [39], log-posterior regularization against frozen pretrained Q-functions [41], supervised observer models [42], and Boltzmann-observer self-rewards in multi-goal teacher–learner pedagogy [40]; goal-recognition design modifies *environments* toward the same early-recognizability objective [52]. Bayesian pedagogy reaches the mechanism from teaching [36, 37], cooperative inverse RL supplies its game-theoretic form, pedagogic equilibria [53], and reward-rational choice unifies behavior-as-evidence [54]; Milli and Dragan analyze, on the learner's side, whether to assume a literal or pedagogic partner [55], the model-choice question we pose on the reward side. Relative to this body of work, our contributions are the Max-Ent/RSA derivation, the first *systematic* listener-choice study in deep MARL under a certified-premium protocol, and the boundary map: the individual listener constructions have precedents, which we measure head-to-head rather than re-invent.

Intrinsic rewards for coordination. Closest to R_{comm} is the auxiliary reward of Tian et al. [45], paying an agent for its partner's correct belief about action-transmitted private information; fixing the audience to the partner's own inference network is one point in the listener space we sweep. We evaluate it head-to-head (Sec. 5.1), alongside a variational instantiation of the conditional-information regularizer $I(A;G|S)$ of Strouse et al. [4]. Social influence [20] rewards *compelling* a partner's reaction; R_{comm} rewards transmission, leaving teammates free to exploit or ignore the revealed intention. In Hanabi, the Bayesian action decoder's public belief lets agents *act informatively* because they know they will be decoded—informativeness from task reward, not rewarded directly [22]; SAD simplifies that machinery [23]; off-belief learning (OBL) grounds interpretation in a non-communicative base policy to keep conventions cross-play-safe [24]; and theory-of-mind intrinsic rewards share private knowledge on the hint channel [56, 57]. Our inverse-planning listener (Sec. 3.2) is the *speaker-side dual* of OBL's grounding move: OBL anchors the listener's belief in a frozen non-communicative policy, we anchor the reward's listener in one. Cross-play evaluations connect to ad-hoc teamwork [44].

Implicit communication. Much MARL research equips agents with an explicit channel [25–27, 43]; we study *implicit* communication [47]: the action is the signal. Sokota et al. [35] encode messages into trajectories via minimum-entropy coupling atop a max-ent policy at a chosen bit-rate; our reward asks for legibility of the task-relevant private state only insofar as it pays, with no coupling machinery at execution.

Computational pragmatics. RSA [2] models communication as recursive speaker–listener reasoning, has a rate-distortion reading [28], and admits learned literal listeners in neural instantiations [29, 30, 48]; RSA-style reasoning has entered emergent communication over explicit channels [46], implicit channels themselves have been learned end-to-end [50], and we run the recursion over *actions*, using its output as a reward. Our protocol contributions continue the measurement skepticism of Lowe et al. [49]—reward-based measures of communication mislead, and positive signaling need not imply positive listening—which the oracle-gap gate, the blind-partner control, and the convergence discipline operationalize. CRSA [3] extends RSA to multi-turn dialog under asymmetric information; Sec. 3.4 locates such purely communicative agents within our composite objective.

3 PRAGMATIC REWARDS

3.1 From RSA to an Intrinsic Reward

The single-step Max-Ent-optimal policy and the RSA speaker are both softmax choice models,

$$\pi^*(a|s) \propto e^{R(s,a)/\alpha_{\text{RL}}}, \qquad S(u|m) \propto e^{\alpha_{\text{RSA}} V_L(u,m)}, \tag{1}$$

identical in the single-step case under $\alpha_{\text{RSA}} = 1/\alpha_{\text{RL}}$, where the listener value $V_L(u,m)$ quantifies the clarity of act u for conveying meaning m [2, 31, 32]. We are explicit about this observation's altitude: it is a structural correspondence, not an equivalence theorem—any Boltzmann-rational model shares the form, the identification is single-step while our agents are sequential, and in practice the two temperatures are decoupled (the SAC entropy coefficient is auto-tuned while the recursion below runs at unit rationality). Its role is to motivate a definition: treating the action as an utterance about the agent's private state m_t (its "meaning," e.g. an intended target), we set

$$R_{\text{comm}}(s_t, a_t) \triangleq V_L(a_t, m_t) = \log L_t^*(m_t|a_t, s_t) + \log |\mathcal{M}|, \tag{2}$$

the *information gain* a pragmatic listener obtains about the true meaning: zero when the action is uninformative, positive as the posterior concentrates. The policy maximizes $R_{\text{total}} = R_{\text{ext}} + \lambda R_{\text{comm}}$.

3.2 The Space of Literal Listeners

Computing L^* requires a *literal listener* $L_0(m|a,s)$. Given L_0 and a set $\mathcal{A}$ of alternatives (the true action plus K sampled ones; uniform unless noted), the pragmatic listener applies N_r RSA steps at unit rationality and uniform prior,

$$S_k(a|m) = \frac{L_{k-1}(m|a,s)}{\sum_{a' \in \mathcal{A}} L_{k-1}(m|a',s)}, \quad L_k(m|a) = \frac{S_k(a|m)}{\sum_{m'} S_k(a|m')}, \tag{3}$$

and $L^* = L_{N_r}$. Because the alternative set is sampled, R_{comm} is stochastic; when alternatives are policy-drawn or L_0 co-adapts, it is additionally policy-dependent, and the Bellman analysis below holds per reward snapshot, as for other co-adaptive intrinsic rewards [13]. We study four hand-crafted instantiations, a co-adapting learned family, and a frozen inverse-planning family.

(i) Direction listeners score actions by cosine similarity to the ideal direction v_m toward the target consistent with m (*simple*), optionally penalizing similarity to competing meanings (*exclusivity*: $s = \alpha_{\text{temp}}(\cos(a, v_m) - \max_{m' \neq m} \cos(a, v_{m'}))$). **(ii) The progress listener** scores the per-step reduction in distance to each candidate target, mirroring cost-to-go legibility [1]. **(iii) The filter listener** maintains a recursive belief $b_t(m) \propto b_{t-1}(m)^{\rho} e^{\alpha_{\text{temp}} \cos(a_t, v_m)}$ with exponential forgetting, rewarding $\log b_t(m_t) + \log|\mathcal{M}|$—explicitly non-Markovian. **(iv) Learned listeners** $L_\theta(m|s,a)$ are trained by maximum likelihood on replayed transitions (private state masked from the input), so the common ground co-adapts and the reward credits only distinctions present in the current behavioral distribution. We also study *bounded* variants restricted to the audience's viewpoint (site bearings and action only) and to the pre-commitment window (Sec. 5.1.1), with $N_r \geq 1$ adding contrastive reasoning on top. **(v) The inverse-planning listener (IPL)** ports the classical Boltzmann goal-inference observer [15, 51]—used as a legibility reward by Faria et al. via per-goal optimal values [16] and as a regularizer against frozen pretrained Q-functions by Persiani and Hellström [41]—to deep MARL with no model, plan, or hand-cost: $L_0(m|s,a) \propto \exp(-\|a - \tanh \mu_{\text{ref}}(s,m)\|^2 / 2\kappa^2)$—an idealized Boltzmann observer of the reference policy, at fixed tolerance κ with the policy's own variance discarded—where π_{ref} is a *frozen, seed-disjoint, non-communicative* goal-conditioned baseline policy (the task's own literal speaker S_0, obtained from self-play). Freezing makes the reward stationary and closes the co-adaptation route to saturation; the semantics are human-readable—"which goal would make this action likely for a plain task-optimal agent?"—and any goal-relevant statistic enters only insofar as the reference policy itself encodes it. Boltzmann-observer self-rewards have reached deep RL in teacher–learner pedagogy [40]; our deltas are the dense per-step reward, the frozen seed-disjoint reference (their inference co-adapts—the saturation-prone choice by Proposition 3), the RSA recursion, and certification under an oracle premium.

3.3 Two Propositions

Proposition 1. *Let meanings be uniform on $\mathcal{M}$ and $L_\theta(m|s,a)$ be any conditional distribution. Under the joint ρ_π induced by the policy,* $\mathbb{E}[\log L_\theta(m|s_t,a_t) + \log|\mathcal{M}|] \leq I(M; S_t, A_t)$*, with equality iff L_θ equals the true posterior ρ_π-almost everywhere. The expected learned-listener reward is the Barber–Agakov bound [33]; ML training tightens it; policy optimization against it maximizes a lower bound on disclosure.*

Proof. $\mathbb{E}[\log L_\theta] = -H(M|S_t,A_t) - \mathbb{E}\,\mathrm{KL}(p(\cdot|s,a) \| L_\theta(\cdot|s,a))$; add $H(M) = \log|\mathcal{M}|$. □

Three remarks. First, the proposition holds for *any* conditional distribution over the same conditioning variables: the instantaneous hand-crafted listeners also yield bounds on the same mutual information, only looser (the filter listener, conditioning on history, bounds the larger $I(M; S_{1:t}, A_{1:t})$). What distinguishes the learned listener is not being a bound but being the *tightest* one. Second, the connections to prior objectives are close but not exact: DIAYN [13] maximizes a bound of this form; Strouse et al. [4] target the conditional $I(A; G|S)$, differing by $I(M; S)$; Eccles et al. [21] shape an explicit channel with entropy targets. All are forms of the literal ($N_r = 0$) level with a learned common ground. Third, the RSA recursion moves L^* away from the ML fit, so it loosens the expected-reward bound whenever that fit is at the posterior—its value, where we find any, is as a *shaping signal*, not an information estimate.

The first proposition also gives the *vocabulary* clause of the audience-matching principle formal content. Model the intended reader as computing its behavior from a feature class Φ of statistics of the observable history (for a feed-forward teammate: functions of raw observations and a short motion window), and say a listener is *vocabulary-restricted to* Φ if it factors as $L(m|s,a) = g(\varphi(s,a))$ for some $\varphi \in \Phi$.

Proposition 2 (Vocabulary restriction). *If L is vocabulary-restricted to Φ, then under the conditions of Proposition 1,* $\mathbb{E}[\log L(m|\varphi(s_t,a_t)) + \log|\mathcal{M}|] \leq I(M; \varphi(S_t,A_t)) \leq I(M; S_t, A_t)$*, i.e., the expected reward lower-bounds the information in the* reader's channel*: policy optimization against it can only pay for codes the reader can extract.*

Proof. Proposition 1 applied to the pushforward channel $(M, \varphi(S,A))$; the second inequality is data processing. □

Two consequences organize the empirical map. First, a vocabulary-restricted listener that *saturates* certifies that the reader-visible channel itself carries the meaning—saturation is then the success condition, not a pathology—whereas an unrestricted listener can saturate through reader-invisible information (context, private regularities), silencing the gradient while communicating nothing the audience can use. Second, our listener family sits at measurable points of this spectrum: the progress and filter listeners factor through motion statistics any observer of raw positions computes (fully restricted); the viewpoint-bounded learned listener restricts φ (slot bearings and the action) but leaves g a free co-adapting network; the full-context learned listener is unrestricted. The empirical ordering of their gains (Sec. 5.1) tracks this restriction ordering exactly.

The second proposition delivers the failure mode that dominates our experiments.

Proposition 3 (Gradient saturation). *Fix a listener L and a policy π_ϕ with joint occupancy ρ_π over (m, s, a), with the reward clipped below at $-B$, and write $B' = B + \log|\mathcal{M}|$ and $g = \nabla_\phi \log \pi_\phi$. If $\rho_\pi(L(m|s, a) \geq 1 - \epsilon) \geq 1 - \delta$, then for the score-function estimator with baseline $b = \mathbb{E}[R_{comm}]$,*

$$\left\|\nabla_\phi \mathbb{E}_{\pi_\phi}[R_{comm}]\right\| \leq \left(-\log(1-\epsilon) + \delta B'\right)\mathbb{E}\|g\| + \sqrt{\delta}\, B' \left(\mathbb{E}\|g\|^2\right)^{1/2}, \tag{4}$$

which is $O(\epsilon + \delta B' + \sqrt{\delta} B')$ per step, scaled by at most $1/(1-\gamma)$ over the horizon.

Proof. Split $\mathbb{E}[|R_{\text{comm}} - b|\, \|g\|]$ over the event $\{L \geq 1 - \epsilon\}$ and its complement. On the good event the reward lies in an interval of width $-\log(1-\epsilon)$, and b lies within $\delta B'$ of that interval (it is a mixture of the good-mass mean and δ-weighted clipped values), so $|R_{\text{comm}} - b| \leq -\log(1-\epsilon) + \delta B'$ there. On the bad event $|R_{\text{comm}} - b| \leq B'$, and $\mathbb{E}[\mathbf{1}_{\text{bad}}\|g\|] \leq \sqrt{\delta}\,(\mathbb{E}\|g\|^2)^{1/2}$ by Cauchy–Schwarz—the bad event may well concentrate where the score is large, so no independence is assumed. □

The high-probability, second-moment form is what our experiments instantiate: Gaussian policies have unbounded support and unbounded scores, so pointwise or independence-assuming versions would be vacuous. And we measure the antecedent directly rather than proxying it with decode accuracy: at convergence the full-context listener holds posterior mass $L(m|s, a) \geq 0.95$ on 93% of occupancy (mean posterior 0.971) and the empirical standard deviation of its R_{comm} is 0.31, while the bounded listeners concentrate less (mean posterior 0.884–0.889, $\mathbb{P}(L \geq .95) \approx 0.6$, reward s.d. 0.34–0.40)—consistent with their residual gradient. (In our implementation the pragmatic branch clamps posteriors at 10^{-8}, giving $B' \approx 19$.) Our agents use reparameterized (SAC) gradients, where the same interval argument bounds the spread of the reward entering critic targets; we therefore use Proposition 3 as the mechanism *consistent with* the observed inertness rather than a quantitative prediction. It formalizes the saturation direction of the audience-matching principle—a listener that decodes everything demands nothing—and the antecedent can be reached two ways: contextual shortcuts that reveal the meaning regardless of the action, or co-adaptation fast enough to decode whatever the current policy emits. We observe both (Sec. 5.1.1).

3.4 Architecture and Objective

A *Pragmatic Reward Module* (PRM) computes R_{comm} offline during centralized training—at collection for static listeners, at replay for co-adapting ones—and never in the action-selection loop; at execution agents act from local observations and the module is discarded (CTDE), except in the explicitly marked *ear* conditions, which deploy the listener's posterior as an extra observation and remain decentralized-executable. The action-value function satisfies the soft Bellman backup applied to R_{total},

$$Q(s_t, a_t) = R_{\text{total}}(s_t, a_t) + \gamma\, \mathbb{E}_{s_{t+1}}\left[\alpha_{\text{RL}} \log \int e^{Q(s_{t+1}, a')/\alpha_{\text{RL}}}\, \mathrm{d}a'\right], \tag{5}$$

with the caveats above (per-snapshot for co-adapting rewards; a Markovian critic absorbs the filter listener's history-dependence as reward noise). Setting $R_{\text{ext}} = 0$, $\lambda = 1$, $\gamma = 0$ collapses Eq. 5 to a soft-max speaker over V_L—the same objective form optimized per turn by dialog models such as CRSA [3]; we state this as a correspondence of objective forms, not a formal reduction.

4 EXPERIMENTAL SETUP

Environments and the oracle-gap criterion. A benchmark for communicative rewards must certify that communication has task value: comparing a baseline against an *oracle* whose observations include all private states measures what intent transmission can unlock. **Environment A: scout-support** (converged oracle gap 0.112/step). A fast *scout* privately knows which of three sites is the target; a slower *supporter* earns a team bonus only while both occupy the target together. The scout must first visit a central *pickup waypoint*, so its first-leg task-optimal motion is identical regardless of target—structurally uninformative—and any early information must be actively signaled by curving the approach. The supporter observes the scout's position, velocity, and a 6-step motion history, never its target; only the scout receives R_{comm}, with a value-of-information weighting (full weight pre-pickup, 0.2 after). Variants: *five meanings* (harder inference), *minefield* (wrong commitment costly, not just slow), *blind* (supporter's view of the scout masked—the causal control), *continuous bearings* ($\mathcal{M} = S^1$; Sec. 5.4), and partner-speed / signal-cost sweeps (Sec. 5.2). **Environment B: preference spread** (oracle gap ≈ 0). A three-agent cooperative-navigation control where proximity resolves coordination, so an oracle gains nothing; it measures what a communicative reward *costs* when there is nothing to say.

Algorithm and conditions. All agents train with multi-agent SAC: parameter-shared Gaussian actors, a centralized twin critic [34] on the scalar team reward with λR_{comm} added, automatic entropy tuning. Conditions differ only in reward and observation content: baseline (λ=0), oracle, the four hand-crafted listeners, the learned family (literal, +RSA, bounded variants), *ear* conditions that append a listener's posterior to the supporter's observation at execution (still decentralized-executable), and two external baselines: *information regularization* in the style of Strouse et al. [4] (reward $\log q_1(m|s, a) - \log q_0(m|s)$ with two decoders, a variational $I(A; M|S)$) and a *partner-belief* reward in the style of Tian et al. [45] (a decoder over the supporter's own observation). Hyperparameter provenance: $\lambda = 0.3$ and the VOI weight were selected in preliminary short-budget sweeps on the standard task, then frozen for every transfer (Appendix A).

Metrics and statistical policy. We evaluate on a fixed 64-episode set (identical layouts across conditions and evaluations; cross-condition tests remain unpaired Welch over seeds); final performance averages the last three evaluations; 400 training cycles (1.28M steps) throughout. *Commitment accuracy*: fraction of steps the supporter's nearest site is the true target. *Post-hoc decodability*: a fresh listener trained from scratch on each converged policy's rollouts. Multiple-comparison policy: headline claims have Welch $t \geq 4$ (surviving Bonferroni across the ~30 tests at α=.05); effects at conventional p<.05 are labeled suggestive and uncorrected. Preregistration here means operationally: the launch specification

Table 1: Environment A at the converged budget (mean ± s.e.m.). Closure is the fraction of the oracle premium closed; Commit is the supporter's commitment accuracy. n=10 seeds unless marked † (n=8) or ‡ (n=6).

Condition	R_{ext}	Closure	Commit
Baseline	0.845 ± .010	—	.851
Oracle	0.957 ± .008	100%	.880
Simple L_0	0.915 ± .010	62%	.857
Exclusivity $L_0^{\dagger}$	0.795 ± .081	n.s.	.850
Progress L_0	**0.924 ± .008**	**70%**	**.861**
Filter L_0	0.920 ± .006	67%	.858
Learned L_θ	0.828 ± .015	−15%	.853
Learned + RSA†	0.863 ± .008	16%	.856
Bounded + RSA (λ=.6)†	0.875 ± .005	27%	.861
IPL, literal†	0.907 ± .014	55%	.864
IPL + RSA (λ=.3)†	0.919 ± .006	66%	.862
IPL + RSA (λ=.6)†	**0.923 ± .009**	**70%**	**.863**
Info-reg. [4]†	0.883 ± .011	34%	.856
Partner belief [45]†	0.852 ± .016	6%	.856
Ear, λ=0‡	0.874 ± .016	26%	.854
Ear + R_{comm}	0.878 ± .012	30%	.858
Filter + ear	0.920 ± .012	66%	.854

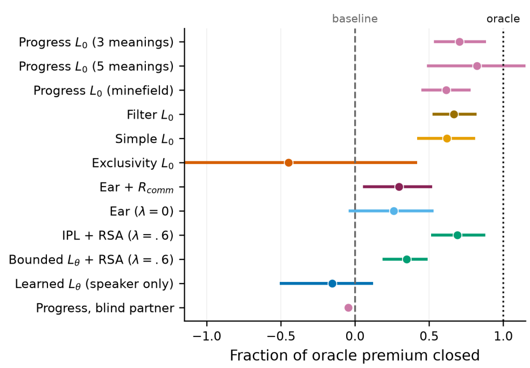

Figure 1: The results on one axis: fraction of each environment's oracle premium closed (dots; bootstrap 95% CIs over seeds). Bounded motion-grounded listeners transfer across environments; the saturated learned listener sits at or below zero; the blind control (scaled by its own premium) sits at zero.

was committed before any run of the cohort started; confirmation cohorts are held-out seeds under a frozen recipe. Per-condition seed counts grew by extension (never selective stopping); no completed run was excluded, and extensions are disclosed where they changed an estimate.

5 RESULTS

5.1 Environment A: Which Listener Pays

Table 1 and Figure 1 summarize Environment A at the converged budget. The converged oracle premium is 0.112 per step (0.957 vs. 0.845; commitment .880 vs. .851)—the irreducible value of early intent information.

Motion-grounded hand-crafted listeners close most of the gap. The strongest conditions are the simplest: progress 70% (Welch $t = 6.2$), filter 67% ($t = 6.5$), plain cosine 62% ($t = 5.0$)—through the speaker-side reward alone, with the largest commitment gains (.857–.861) and visibly more legible scouts (fixed geometric-probe accuracy .61 vs. .54; the fresh-decoder time profile in Fig. 2 confirms the separation). These listeners credit only *overtly* geometric signals, and overt signals are exactly what the supporter's own policy network can read from raw observations; the channel is the behavior itself. The exception proves the calibration rule: the exclusivity listener, whose margin demand is geometrically unsatisfiable, is unstable across seeds (0.795 ± .081; a 1-in-8 collapse rate: seven seeds in 0.84–0.93 and one collapsed at 0.23—an early n=3 estimate of +16% closure was eliminated by extension to n=8).

A blind-partner control certifies the mechanism. Masking the supporter's view of the scout (8 seeds; blind oracle 4), the pragmatic gain inverts into a significant cost (−1.67 vs. −1.56; $t = 2.8$, $p \approx .02$, uncorrected)—the scout still pays the signaling detour for messages nobody receives—while the blind *oracle* retains full performance (0.963). The effect is communication, not shaping.

A too-competent listener exerts no pressure. Trained to convergence, the full-context L_θ decodes the scout at 97% *whatever the scout does*: its reward saturates near the log 3 ceiling, and the speaker-side learned condition ends below baseline (0.828, −15%). Deployed instead as the supporter's *ear*, the same listener recovers 30% ($t = 2.15$, $p \approx .05$, suggestive): with a saturated listener, the value is in the decoding, none in the speaking. Combining a hand-crafted reward with its ear matches the reward alone—once the speaker signals overtly, the equipped ear is redundant.

External baselines, head-to-head. We run the two closest prior reward designs under the frozen recipe (λ=0.3, VOI, 8 seeds, 400 cycles). The partner-belief reward in the style of Tian et al. [45]—a decoder over the supporter's own observation—lands at baseline parity (0.852 ± .016, 6% closure, n.s.), with a small commitment gain (.856): fixing the audience to the actual partner's information state is exactly the saturating choice our analysis warns against. The information-regularization reward in the style of Strouse et al. [4]—$\log q_1(m|s,a) - \log q_0(m|s)$—closes 34% (0.883 ± .011, $t = 2.6$, suggestive), statistically indistinguishable from our bounded+RSA recipe, and for a principled reason: subtracting the state decoder removes the contextual shortcut *by construction*, implementing the viewpoint bound by subtraction where RSA implements it by contrast. Tuning is equalized: each design received the same weight sweep (three points for information regularization, two for partner belief), and information regularization *peaks* at λ=0.3 (falling to 3% at λ=0.6 and 13% at λ=1.0, both n.s.) exactly where the recursion reward peaks at λ=0.6—the two context-debiasing routes have opposite weight tolerance. The two reward designs, run under a common harness,

Table 2: Bounding the learned listener (8 seeds per condition, except the full-context and progress rows, which repeat Table 1's n=10 entries; premium 0.112; λ=0.3 unless noted). "Sat." is converged decode accuracy. The cleanest controlled contrast (suggestive) is the λ=0.6 block: recursion vs. no recursion at the same weight is worth 0.064 ($t \approx 3.85$; the n=16 row pools the selection cohort with the held-out-seed confirmation and is secondary to the confirmation-only estimate in the text).

Condition	R_{ext}	Closure	Commit	Sat.
Learned (full context)	0.828 ± .015	−15%	.853	.97
+ viewpoint bound	0.849 ± .017	4%	.851	.95
+ viewpoint, + RSA	0.868 ± .011	21%	.856	.95
+ window bound	0.841 ± .017	−4%	.853	.94
+ window, + RSA	0.868 ± .009	21%	.856	.94
Window bound, λ=0.6	0.820 ± .016	−22%	.854	.94
+ RSA, λ=0.6 (n=16)	**0.884 ± .006**	**35%**	**.861**	.94
+ RSA, λ=1.0	0.849 ± .013	4%	.864	.96
Progress (hand-crafted)	0.924 ± .008	70%	.861	—

thus occupy the same band at their respective optima—and both trail the vocabulary-restricted hand-crafted listeners by a factor of two, which is the audience-matching principle's second clause made quantitative. (The partner-belief design remains inert at every weight tried: 0.833–0.852.)

5.1.1 Bounding the Learned Listener. If the learned listener fails because it is too strong, audience matching predicts a remedy: weaken it toward the intended reader. A *viewpoint-bounded* listener sees only the site bearings and the action—what the audience uses to interpret the utterance—removing the contextual shortcut; a *window-bounded* variant is additionally trained only on pre-commitment steps, so decode accuracy cannot be earned from the post-commitment leg (Table 2).

The first result is a negative control that sharpens the principle. Both bounded listeners *re-saturate*, later (decode accuracy 0.54 at quarter-training for window-bounded vs. 0.88 full-context) and by a different route, co-adapting to whatever the current policy emits, the second saturation mode of Proposition 3. Their literal rewards end at baseline parity, curing the harm but recovering nothing. Layering RSA recursion on top lands at 0.868 (21% closure) in two replications, ahead of its literal counterpart at all but one evaluation.

The second result is the paper's cleanest causal contrast. Raising the weight (λ=0.6, the middle of three swept) lifted the recursed variant to 0.893 ± .009 (43%) on the selection runs; a confirmation on eight held-out seeds then showed the expected regression to the mean (Sec. 5.2.1). Because the selection cohort chose λ, the confirmation cohort is primary: 0.875 ± .005, a 27% closure (t = 2.7); pooling both cohorts gives 0.884 ± .006 (35%, t = 3.4) but re-imports the selection bias (secondary). Commitment accuracy is .861 in both cohorts, matching the progress listener. The control is decisive: the *literal* listener at the same weight falls *below* baseline (0.820 ± .016), since amplifying a saturated reward amplifies noise and cost, not signal. Recursion vs. no recursion at matched weight is worth 0.064 ($t \approx 3.9$), suggestive and replicated in direction across four cohorts. Two further levers are inert (policy-drawn alternatives 0.872; a 10× slower listener 0.861). The deployed-ear variant is redundant by construction and lands at baseline parity (0.841 ± .023; an earlier catastrophic estimate was an evaluation bug disclosed in the appendix).

Why the residual gap to the hand-crafted family (27% vs. 70%)? On commitment the recursed listener has fully caught up (.8606 ± .0007, exactly matching progress; the gain over baseline's .8508 ± .0013 is $t \approx 6.5$, headline-grade on its own), but the co-adapting learned listener remains satisfied by *any* decodable regularity, including idiosyncratic codes the co-training supporter tracks imperfectly.

The inverse-planning listener closes the gap. The IPL (listener (v)) resolves the residual outright, with no hand-crafted geometry: its literal form closes 55% (0.907 ± .014, t = 3.6), and with one RSA step it reaches 66% at λ=0.3 (0.919 ± .006, t = 6.3) and **70%** at λ=0.6 (0.923 ± .009, t = 5.8)—*equal to the progress listener*, with commitment accuracy .862–.864 exceeding it. Both recursed points clear the $t \geq 4$ headline bar—the first learned-family results in this study to do so—and the two-point λ robustness (66%/70%) is a plateau, not a selected optimum: the two λ points are the only IPL runs made, and no selection/confirmation split was needed since neither was chosen post hoc. The result satisfies both clauses of the audience-matching principle *as instantiated*: frozen, hence non-saturating—its reward cannot chase the policy (clause a, Proposition 3)—and anchored in task-competent motion, reader-visible features (clause b, Proposition 2). What the co-adapting bounded listeners lacked was never hand-crafted geometry; it was a *non-co-adapting, competence-grounded* literal listener. The constructive recipe for any goal-conditioned task: train it without communication, freeze the policy, Bayes-invert it at soft rationality, add the pragmatic recursion.

One recipe, three task variants; robustness. The winning recipe (progress, λ=0.3, VOI, frozen) transfers untuned to two structurally distinct variants of the same environment family: five meanings closes 82% (t = 3.98, p = .0014, surviving Bonferroni), minefield 61% (t = 5.2, p < .001)—majority-gap closure in both. The recipe is a plateau, not a tuned point: 0.894–0.925 across the $\lambda \in \{0.1, 0.3\}\times$VOI grid (low end at λ=0.1, uniform weighting; 3 seeds/cell), with a cliff at λ=0.6 under VOI weighting (0.450 ± .203; 0.823 uniform)—the same λ that is the *optimum* for the bounded learned-pragmatic reward: the safe weight range scales inversely with reward magnitude. Zero-shot cross-play retains 92–94% of self-play reward (no worse than baseline's cross-seed transfer); at 16× parameters the qualitative ordering is unchanged though the premium shrinks—readers strengthen.

5.2 When Does Pragmatism Pay? A Boundary Map

We vary the two quantities that should govern the effect—how much early intent is worth (partner speed 0.45–0.90 sets the oracle premium) and what signaling costs (scout shaping weight 2×, 4×)—5 seeds per setting, everything else frozen, and plot every setting on premium-vs-gain axes (Fig. 3).

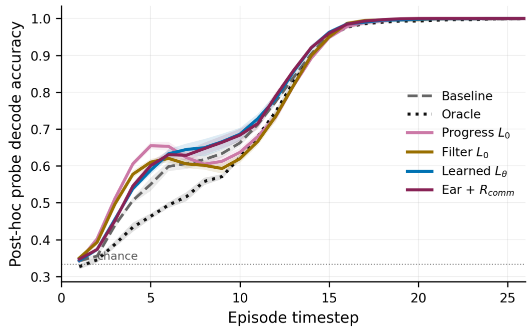

Figure 2: Per-timestep decode accuracy of a fresh post-hoc probe on converged behavior. In the early decision window the progress-listener scouts carry the most decodable intent while oracle scouts—whose teams never rely on motion—are the *least* legible early: legibility is bought exactly when the partner is still deciding.

An empirical regularity, stated with its scope. Wherever the premium is resolvable (point estimate > 3× its s.e., fixed before analysis), the recipe captures a roughly constant fraction: an origin-constrained fit over all seven resolvable settings gives gain ≈ 0.51× premium (bootstrap 95% CI [0.42, 0.59]). Two caveats are explicit. The two failure-mode settings below (ratio 0.37) have the *largest* premiums and dominate the fit; excluding them, the remaining five give 0.68× (CI [0.57, 0.79], ratios 0.59–0.82)—but that classification is post hoc, so 0.68 is exploratory and 0.51 primary. And gain and premium share each setting's baseline estimate, so their errors are correlated; the bootstrap resamples seeds within settings. All settings come from one environment family and one algorithm: this is a regularity of that family, not a law, and would be falsified by a resolvable-premium, healthy-channel environment with capture outside the fitted range.

Failure modes. Where no premium is resolvable, no gain appears (preference spread t=0.2 at the converged budget; 4× cost t=1.4). Where the premium is real but the partner too slow (0.45) or signaling too expensive (2×), capture drops to 0.37×: the premium is partly unreachable at any signaling latency, or the scout rationally emits less. Where the channel is severed (blind), the reward does not merely stop helping—it *hurts*, because the speaker pays for signals nobody receives. Table 3 consolidates the map as a checklist; each row is a way we made the method fail.

The boundary is two-sided: something to say, and not already said. A highway on-ramp negotiation we built as a candidate third environment closed the boundary from the far side. There the merger's *task-optimal* approach—aligning with its intended gap—is inherently informative, and baseline RL bootstrapped meaning-differentiated accommodation unaided: listeners braked selectively for the correct gap under no communicative reward, leaving the oracle premium unresolvable (0.021 ≈ 1.3× s.e.) even though severing the channel cost 26 points of success. *Signal pre-existence* is thus the mirror of Environment B's failure mode: legibility rewards can only buy value when the agent has something to say (else they purchase certified

Table 3: When pragmatism pays. Each row is a condition varied to failure; "gain" is relative to the matched baseline.

Requirement	Violated by	Gain
Premium exists	Pref. spread; 4× cost	n.s.
Partner observes speaker	Blind partner	−0.11
Partner can act in time	Partner speed 0.45	0.37×
Signal is affordable	2× shaping	0.37×
Listener credits readable code	Literal L_θ, any λ	n.s.
Listener is calibrated	Exclusivity L_0	n.s.
Weight matches reward scale	$\lambda \geq 0.6$	≤ 0
Evaluated at convergence	150 cycles	flips
Meanings coarsely separable	S^1 continuum	0.18×
Task does not pre-carry signal	On-ramp merge	0.02
All hold	*3 task variants*	.59–.82×

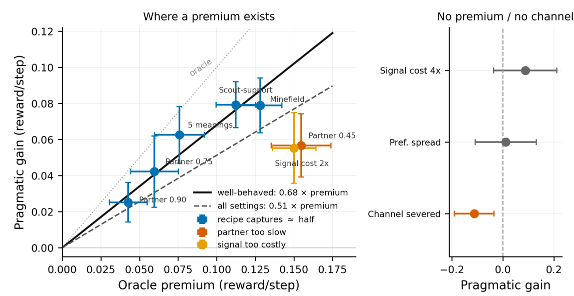

Figure 3: The boundary map. Left: across every setting with a resolvable premium, gain scales with premium (solid: well-behaved fit 0.68×; dashed: all settings 0.51×; dotted: oracle). Settings where the partner is too slow or signaling too costly fall below despite large premiums. Right: no premium / no channel. Left bars ±1 s.e.m., right bars 95% CIs.

miscalibration, Sec. 5.3) *and* the task does not already say it (else optimization substitutes for communication). The 0.51× regularity quantifies the region between—noting its pre-registered scope is the partner-speed/cost family: the meaning-space regimes of Sec. 5.4 fall below its fitted band (capture 0.18–0.26), read as the axis along which certification weakens, not as a violation of the within-family fit.

5.2.1 The Pre-Convergence Hazard. This benchmark produced three successive spurious conclusions before the converged one: evaluated at increasing budgets (Fig. 4), the progress listener's effect reads as strongly negative through 0.5M steps (≈ −60% closure at our early budget) and +70% only past ∼1M, and at 150 of the ∼400 needed cycles hand-crafted listeners appeared destructive while an ear condition appeared to close 30% of a then-larger gap—evaporating when seeds were doubled. Communicative-MARL comparisons are unusually exposed because the quantity of interest is a co-adaptation between policies and listeners, converging more slowly than either alone; we recommend reporting training-stage sensitivity in the form of Fig. 4.

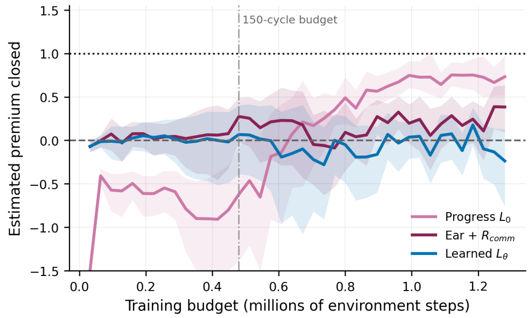

Figure 4: The pre-convergence hazard: estimated fraction of the premium closed vs. the training budget at which the comparison is made (bootstrap 95% bands, 10 seeds). The ordering stabilizes only past ~1M steps.

5.3 Environment B: The Cost of Signaling with Nothing to Say

Environment B measures what a communicative reward costs when there is nothing to say (converged 400-cycle budget, 5 seeds; full table and figures in Appendix B). The oracle confirms the near-zero premium (1.93 vs. 1.86, +0.07 ± .08, n.s.; an earlier 150-cycle evaluation put the oracle *below* baseline—the pre-convergence hazard of Sec. 5.2.1, which also inflated the miscalibration tax below from 9% to 46%). Calibrated learned listeners are a free add-on: task reward at baseline (1.93 ± .06 literal, 1.87 ± .02 recursed), communicative reward at genuinely *positive* information gain (+0.21/+0.22), the recursion again extracting more decodable signal than literal decoding (listener accuracy .681 vs. .655). Miscalibration is a tax with a free diagnostic: the exclusivity listener's own reward is stuck *below* the uninformative level (−1.83; its margin demand is geometrically unsatisfiable) and it costs 9% of task reward (1.69, t = 2.3, suggestive). A post-hoc coda sharpens the diagnosis: that condition's behavior is the *most* decodable of all to a fresh probe (.700 ± .008 vs. .665 ± .007, t = 3.2, uncorrected), yet its own listener never credits it—miscalibration is a vocabulary mismatch, not an absence of signal, and a reward persistently below its uninformative level certifies it at zero measurement cost.

5.4 The Meaning-Space Axis: Discrete, Continuous, Self-Assigned

The map's final axis varies the meaning space itself. Its regimes are not evidentially equivalent—discrete-meaning closures meet the pre-registered headline bar (t≥5.8) while continuous and self-assigned closures are suggestive (t=2.2–2.3)—and we present them as one axis to chart where certification weakens, not to claim uniform strength. One ordinal pattern does hold throughout: the literal listener is inert everywhere, and the recursive listener's advantage never reverses sign; its magnitude is decisive only in the discrete regime (Fig. 5).

5.4.1 Continuous Meanings: the Particle Estimator. Everything so far assumed a small discrete meaning space. What breaks on a

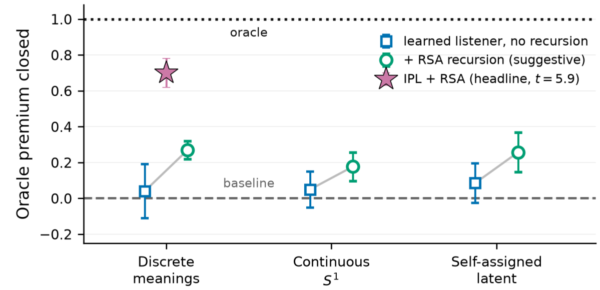

Figure 5: The meaning-space axis. In each regime the *learned* listener without recursion sits near the baseline; one RSA step lifts it by a similar fraction everywhere, but only the inverse-planning listener in the discrete regime clears the t≥4 headline bar (filled star). The hand-crafted listeners (Table 1, 60–82%) are a separate family, omitted here to isolate the recursion's effect on the learned one. Whiskers are ±1 s.e.m. as a fraction of each regime's oracle premium.

continuum is not the literal decoder but the recursion: RSA's listener normalization integrates over $\mathcal{M}$. We replace it with *contrast particles*: the true meaning is scored against K=15 alternatives drawn from other episodes in the batch, both RSA normalizations become finite softmaxes, and the recursion becomes alternating row/column normalization of the (alternatives × particles) score matrix—steps of the Sinkhorn scaling that Wang et al. identify as the mathematical form of cooperative-communication recursions, RSA included [10]. Our contribution is the estimator, not the identification: in-batch particles turn the correspondence into a trainable reward on continuous meaning spaces, and at zero recursion depth the objective is *exactly* InfoNCE [11] at the true meaning, so the reward inherits the contrastive $\log(K{+}1)$ ceiling—the $\log|\mathcal{M}|$ analog—and Propositions 1 and 3 carry over with the InfoNCE bound; sampled-alternative RSA is familiar from pragmatic captioning [12]. One design is excluded on principle rather than results: a density decoder $q(m|s, a)$ rewarded by its log-density admits variance collapse—reward inflating without new information, the unbounded analog of saturation—so the contrastive ratio form is a requirement, not a convenience.

We test on a continuous-bearing scout-support: the target is a point at angle $\theta \sim U[0, 2\pi)$ on the site circle, all other mechanics unchanged. The premium re-certifies (baseline 0.843 ± .007, oracle 0.945 ± .008, n=8; blind commitment at chance), and the results are uniformly humbler than their discrete counterparts. The literal contrastive listener is inert (4.9% closure, t=0.5) while its decoder concentrates (top-1 accuracy .74 against a .06 chance level)—the saturation signature again. One Sinkhorn iteration helps directionally in both cohorts of a pre-registered split—24.2% (t=2.6) on the selection seeds, 11.2% (t=1.2) on the confirmation seeds, 17.7% pooled (n=16, t=2.2, suggestive)—with a flat λ response (23.0% at λ=.6, selection cohort). The hand-crafted progress listener, ported for free by scoring particles, drops from 70% closure (discrete) to 15.8% (t=1.6, n.s.). The recursion-over-literal ordering thus replicates in a third setting, but nothing on S^1 clears our headline bar: when the meaning space is dense, nearby meanings have near-identical action consequences, contrast sets lose resolution, and *every* listener family captures less. We read this as a boundary-map extension (Table 3): legibility rewards need meanings coarse enough that alternatives are behaviorally discriminable.

5.4.2 Self-Assigned Meanings: Legible Latents. The continuum still assumed the meaning is given. The last step drops that too: the meaning becomes a latent the agent itself computes. The scout's private observation—enriched with two decoy bearings and two noise channels, only the true bearing decision-relevant—now reaches its policy only through a variational encoder (four dimensions, KL to $\mathcal{N}(0, I)$ at β=10^{-3}) whose output z is concatenated with the public observation to drive the action head. The wiring is InfoBot's [5], sign-flipped: InfoBot *minimizes* goal information in behavior; we keep its KL—bounding $I(Z; \text{private})$—and add a listener bonus on the action channel. The combination, not the encoder, is what is new here, and the reward mechanism itself has a free-latent ancestor in Any-Play's teammate-side skill discriminator [6] (Table 4). Because z is the only route from private information to behavior, what it carries is decided by task pressure and compression—not by us—and R_{comm} becomes the particle recursion of Sec. 5.4 aimed at the speaker's own per-episode posterior mean μ_z: make the representation that drives your decisions readable from how you act. All conditions share the architecture and β, so the anchors price communication, not capacity.

What this regime certifies is the premium. An oracle advantage exists for a latent whose content the designer never specified (bottleneck baseline 1.001 ± .006, oracle 1.089 ± .013, n=8, 6× s.e.; blind commitment near chance, .373 vs. 1/3). The recursion's capture of that premium is suggestive-tier. The literal contrastive reward on μ_z is inert (8.5%, t=0.8), while one Sinkhorn iteration closes 25.7% (t=2.3 at λ=.3; 23.0% at λ=.6), with commitment .890–.895 against .883/.907 anchors. The paper's ordering thus replicates a fourth time, now with no hand-specified meaning anywhere in the loop.

Post-hoc probes explain the mechanism (Fig. 6). For content, the true bearing is linearly decodable from z at R^2=.999 in every condition, the decoys only at ≈.21 and the noise channels partially (.45–.52): decision-relevant content dominates the latent without being specified. For transparency we use matched InfoNCE critics, with K=255 on the private side (ceiling $\log 256 \approx 5.55$ nats) so the estimate is not ceiling-pinned. $I(Z; \text{private})$ is constant across conditions (5.16 to 5.18 nats) while $I(Z; \text{behavior})$ rises from 1.755 ± .007 to 1.782 to 1.818±.012 (baseline, literal, recursed). The transparency coefficient $I(Z; \text{behavior})/I(Z; \text{private})$ climbs .340 to .344 to .351 (baseline vs. recursed t=3.7, post hoc and hence suggestive). The reward made behavior expose the latent while touching neither its content nor its information budget: compression and task pressure choose *what* enters z, and the pragmatic reward chooses *how much of it shows.* A compression sweep separates the two dials. Raising β to 3×10^{-2} drives the budget from 10.6 to 3.1 nats and collapses noise leakage (R^2 from .49 to .11) while the true bearing stays decodable (≥ .97), and the readability gap persists at every β (task gains unresolved at n=4 per point).

We call such representations *legible latents.* Table 4 places the claim among its relatives. Prior exposure-direction rewards target *given* private variables [4, 9, 19, 45, 56], and prior latent-discriminability rewards target *free* skill noise [6, 13]. The nearest relative is VQ-VIB [7], a learned information-bearing latent read by a teammate, but over an explicit channel and with no behavior-legibility reward or certification. To our knowledge, the certified premium for an information-bearing learned latent, and the content-versus-exposure dissociation that verifies it (cf. amnesic probing [8]), appear here first.

Table 4: The self-assigned regime against its nearest relatives. The third column is this paper's own protocol; prior work predates it.

	z learned, info-bearing	Teammate decoder	Certified premium
DIAYN family [13]	no (free)	no	no
Any-Play [6]	no (free)	yes	no
InfoBot [5]	yes (hidden)	no	no
Strouse [4]	no (given)	no	no
Caselles-Dupré [40]	no (given)	no	no
Tian [45]	no (given)	yes	no
MAAL [19]	no (given)	yes	no
VQ-VIB [7]	yes	yes	no
This work	yes	yes	yes

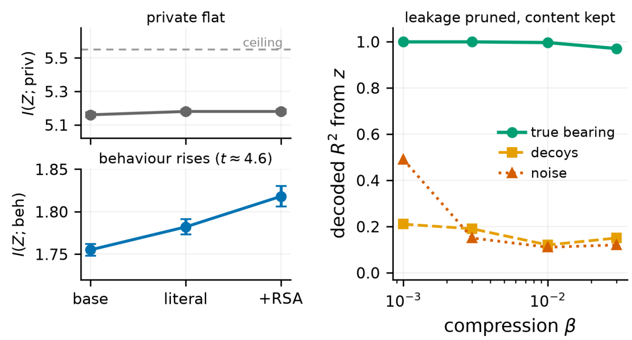

Figure 6: Two separable dials in the self-assigned regime. Left (each strip on its own scale): the legibility reward raises $I(Z; \text{behaviour})$ (t≈4.6) while $I(Z; \text{private})$ stays flat, well below the K=255 critic ceiling. Right: stronger compression (β) prunes decoded noise leakage from z while the true bearing stays fully decodable.

6 DISCUSSION

The audience-matching principle, sharpened by its own ablation. The learned listener decodes the scout at 97% regardless of behavior, so its reward saturates and, by Proposition 3, the shaping gradient vanishes. Bounding it to the audience's viewpoint and decision window is necessary but not sufficient: a bounded co-adapting listener still saturates, chasing whatever code the current policy emits. What separates the crude geometric listeners is what they can credit, namely overtly geometric motion that a feed-forward teammate reading raw observations can also exploit.

An omniscient audience needs no speech, and an infinitely accommodating one accepts any private code. The principle is therefore not "model your audience accurately". The maximum-likelihood listener is provably the *most* accurate model of what behavior reveals (Proposition 1), the partner-belief reward uses the true audience itself, and both are inert: the best model of the audience is the worst reward. Both clauses now have the formal backing the propositions deliver: anchor R_{comm} in a listener that (a) stays non-saturating under the training distribution (Proposition 3) and (b) factors through reader-visible statistics (Proposition 2). What survives learning is RSA's recursion, which converts an inert decoder into a reward recovering a quarter of the premium: the converged posterior no longer varies with behavior, but the contrast against foregone alternatives still does.

Calibration and shaping pressure trade off. Training L_θ on the agents' own data guarantees the common ground never demands what the policy cannot express, which is why it optimizes smoothly and costs nothing where there is nothing to say (Environment B). The same property seeds the saturation failure where there *is* something to say: a co-adapting listener converges toward the posterior, and the closer it gets, the less its log-posterior varies with behavior.

Communication is a two-sided market. The overt-signal route recovers about twice as much as equipping the receiver with an execution-time ear (Table 1); when the intended reader is weak, the burden of clarity belongs on the speaker.

Limitations. Intrinsic rewards densify learning signals, and commitment-timing and probe metrics mitigate but do not eliminate the confound. The audience-matching principle is demonstrated for one audience class (feed-forward policies over raw observations), one environment family, and one algorithm; how listener competence should scale with reader competence is open, and the wide-network check suggests the premium itself shrinks as readers strengthen. Meanings are hand-specified in the headline results. Sections 5.4–5.4.2 relax discreteness and then hand-specification, but closures there fall to the suggestive band, and both the meaning-space geometries that support legibility rewards and a pre-registered second-environment replication of the self-assigned regime remain open. Validation is confined to AI–AI teams; the principle predicts that rewards anchored in human-competence listeners should transfer best to human-agent teams. Inverting λ's sign turns legibility into obfuscation, suggesting privacy applications [4].

7 CONCLUSION

We connected pragmatic inference and maximum-entropy RL, importing the RSA listener value as an intrinsic reward for legible behavior whose decisive design choice is the literal listener. Where communication has no task value, miscalibrated listeners charge for undecodable signaling while calibrated ones approach a no-op. Where it is certifiably valuable, crude motion-grounded listeners that credit only overt signals recover 61–82% of the oracle premium, while a listener competent enough to decode anything demands nothing. That failure, which our saturation bound explains and the measured antecedent supports, is resolved constructively: bounding the co-adapting learned listener cures its harm but not its inertness, and the *inverse-planning listener*, a classical Boltzmann observer self-instantiated from a frozen non-communicative baseline and sharpened by one step of RSA recursion, closes 70% of the premium with no task-specific geometry while remaining human-readable. The audience-matching principle is RSA's oldest lesson restated for reward design: speech is optimized against a literal listener because that listener is limited, in competence and in what it will hear. The recipe is general: train the task without communication, freeze the policy, Bayes-invert it, and let pragmatics do the rest. The construction also reaches past given meanings, since routing the latent through a bottleneck makes an agent's own decision variable legible, so the reward's meaning space can be discovered rather than specified. Agents whose competence includes being understood must be trained against audiences that need the help.

A HYPERPARAMETERS AND REPRODUCIBILITY

Table 5: Hyperparameters (all conditions unless noted).

Parameter	Value
Parallel environments / episode length	64 / 50
Training cycles (both envs.; Env. B λ-sweep)	400 (150)
Actor / critic architecture	MLP 256 × 256
Learning rate (actor, critic) / listener	3×10^{-4} / 10^{-3}
Discount / target smoothing	0.95 / 0.005
Batch / updates per env step	512 / 4
Entropy target	$-\dim(\mathcal{A})$ (auto)
Listener L_θ	MLP 128 × 128
Direction listener temp. / K / N_r	5 / 64 / 1
Learned+RSA alternatives / N_r	16 / 1
Filter temp. / forgetting ρ	2 / 0.9
λ (Env. B / Env. A) / VOI weight	0.1 / 0.3 / 0.2
Scout / supporter speed; hist. window	1.0 / 0.6; 6
Bonus / cover / pickup radius	3.0 / 0.3 / 0.3
Integration step / damping / accel. gain	0.1 / 0.25 / 5.0
Arena clamp / site jitter / waypoint jitter	±1.5 / 0.1 / ±0.25
Mine penalty; replay; warm-up	1.5; 400k; 2000
Evaluation	every 10 cyc. (64 eps.)

All environments, training, aggregation, and figure code will be released, with launch specifications determining every Environment A run exactly (JSON specs including seeds) and the Environment B command line documented in the repository README. One disclosed bug: an earlier snapshot evaluated the bounded-listener ear condition without the posterior injection it was trained with, producing an artifactual 0.551 that persists in the archived histories; the corrected evaluation (0.841 ± .023), reproducible from the released checkpoints via the included re-evaluation script, is what we report.

B ENVIRONMENT B DETAILS

Three agents and three colored landmarks in a continuous arena; each agent holds a private color preference; the team is rewarded

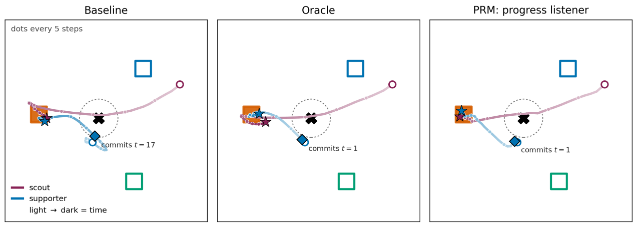

Figure 7: One Environment A evaluation episode under three conditions (selected from 60 candidates to maximize the baseline-vs-recipe commitment-time difference; statistics are in Table 1). Squares are sites (target filled); cross and dashed circle mark the pickup waypoint; trajectories darken with time; diamond = the supporter's commitment moment. The baseline supporter commits at t=17; the pragmatically rewarded supporter commits at t=1, matching the oracle.

for matched coverage, penalized for distance and collisions. Meanings are the three colors; every agent receives R_{comm} under its own listener; $\lambda = 0.1$.

Table 6: Environment B at the converged budget (400 cycles; mean ± s.e.m., 5 seeds). R_{comm} on the information-gain scale (0 = uninformative); List. acc. is the co-trained listener's converged accuracy.

Condition	R_{ext}	Spec.	R_{comm}	List.	Post-hoc
Baseline	1.86 ± .06	.859	—	—	.665
Oracle	1.93 ± .06	.862	—	—	.672
Simple L_0	1.81 ± .05	.849	−0.92	—	.684
Exclusivity L_0	1.69 ± .05	.832	−1.83	—	.700
Learned L_θ	**1.93 ± .06**	.847	+0.21	.655	.672
Learned + RSA	1.87 ± .02	.837	**+0.22**	**.681**	.684

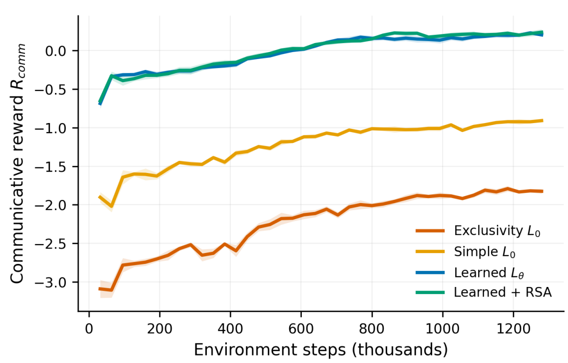

Figure 8: Achieved communicative reward R_{comm} over training (each condition under its own listener; 0 = uninformative). Calibration levels separate: learned listeners cross into positive information gain; the exclusivity listener never leaves negative territory—the miscalibration certificate.

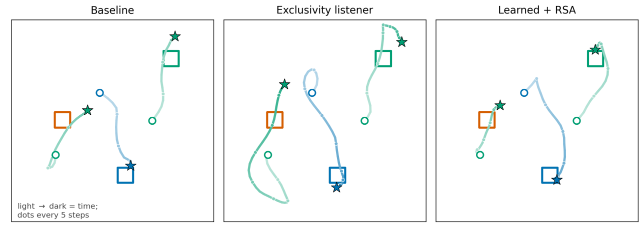

Figure 9: A converged evaluation episode under three conditions. Squares are landmarks (color = category); trajectories are colored by the agent's private preference and darken with time. The exclusivity-listener agents sweep the arena and settle off-landmark—costly motion signaling nothing; the learned-pragmatic agents take direct, mutually distinct paths.

The weight λ (exploratory, 150 cycles, 2 seeds/point). With the learned listener, λ behaves like a dial along an information–performance frontier, up to genuinely positive information gain at a large task cost at $\lambda = 1$; with the exclusivity listener no operating point buys information—the reward is ignored at small λ and destructive at moderate λ. Calibration is what turns λ from a landmine into a dial.

REFERENCES

[1] Anca D. Dragan, Kenton C.T. Lee, and Siddhartha S. Srinivasa. Legibility and predictability of robot motion. In *HRI*, 2013.
[2] Noah D. Goodman and Michael C. Frank. Pragmatic language interpretation as probabilistic inference. *Trends in Cognitive Sciences*, 20(11):818–829, 2016.
[3] Lautaro Estienne, Gabriel Ben Zenou, Nona Naderi, Jackie Chi Kit Cheung, and Pablo Piantanida. Collaborative rational speech act: Pragmatic reasoning for multi-turn dialog. In *EMNLP*, 2025.
[4] DJ Strouse, Max Kleiman-Weiner, Josh Tenenbaum, Matt Botvinick, and David J. Schwab. Learning to share and hide intentions using information regularization. In *NeurIPS*, 2018.
[5] Anirudh Goyal, Riashat Islam, DJ Strouse, Zafarali Ahmed, Matthew Botvinick, Hugo Larochelle, Yoshua Bengio, and Sergey Levine. InfoBot: Transfer and exploration via the information bottleneck. In *ICLR*, 2019.
[6] Keane Lucas and Ross E. Allen. Any-Play: An intrinsic augmentation for zero-shot coordination. In *AAMAS*, 2022.
[7] Mycal Tucker, Roger Levy, Julie Shah, and Noga Zaslavsky. Trading off utility, informativeness, and complexity in emergent communication. In *NeurIPS*, 2022.
[8] Yanai Elazar, Shauli Ravfogel, Alon Jacovi, and Yoav Goldberg. Amnesic probing: Behavioral explanation with amnesic counterfactuals. *TACL*, 9:160–175, 2021.
[9] Miguel Faria, Francisco S. Melo, and Ana Paiva. "Teammates, am I clear?": Analysing legible behaviours in teams. *arXiv:2507.21631*, 2025.
[10] Pei Wang, Junqi Wang, Pushpi Paranamana, and Patrick Shafto. A mathematical theory of cooperative communication. In *NeurIPS*, 2020.
[11] Aäron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv:1807.03748*, 2018.
[12] Reuben Cohn-Gordon, Noah Goodman, and Christopher Potts. Pragmatically informative image captioning with character-level inference. In *NAACL*, 2018.
[13] Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. Diversity is all you need: Learning skills without a reward function. In *ICLR*, 2019.
[14] Brian D. Ziebart, Andrew L. Maas, J. Andrew Bagnell, and Anind K. Dey. Maximum entropy inverse reinforcement learning. In *AAAI*, 2008.
[15] Chris L. Baker, Julian Jara-Ettinger, Rebecca Saxe, and Joshua B. Tenenbaum. Rational quantitative attribution of beliefs, desires and percepts in human mentalizing. *Nature Human Behaviour*, 1(4):0064, 2017.
[16] Miguel Faria, Francisco S. Melo, and Ana Paiva. "Guess what I'm doing": Extending legibility to sequential decision tasks. *Artificial Intelligence*, 330:104107, 2024.
[17] Shuwa Miura and Shlomo Zilberstein. A unifying framework for observer-aware planning and its complexity. In *UAI*, 2021.
[18] Shuwa Miura and Shlomo Zilberstein. Observer-aware planning with implicit and explicit communication. In *AAMAS*, 2024.

[19] Yanyu Liu, Yinghui Pan, Yifeng Zeng, Biyang Ma, and Prashant Doshi. Active legibility in multiagent reinforcement learning. *Artificial Intelligence*, 346:104357, 2025.
[20] Natasha Jaques, Angeliki Lazaridou, Edward Hughes, Caglar Gulcehre, Pedro A. Ortega, DJ Strouse, Joel Z. Leibo, and Nando de Freitas. Social influence as intrinsic motivation for multi-agent deep reinforcement learning. In *ICML*, 2019.
[21] Tom Eccles, Yoram Bachrach, Guy Lever, Angeliki Lazaridou, and Thore Graepel. Biases for emergent communication in multi-agent reinforcement learning. In *NeurIPS*, 2019.
[22] Jakob N. Foerster, Francis Song, Edward Hughes, Neil Burch, Iain Dunning, Shimon Whiteson, Matthew Botvinick, and Michael Bowling. Bayesian action decoder for deep multi-agent reinforcement learning. In *ICML*, 2019.
[23] Hengyuan Hu and Jakob N. Foerster. Simplified action decoder for deep multi-agent reinforcement learning. In *ICLR*, 2020.
[24] Hengyuan Hu, Adam Lerer, Brandon Cui, Luis Pineda, Noam Brown, and Jakob Foerster. Off-belief learning. In *ICML*, 2021.
[25] Jakob Foerster, Ioannis Alexandros Assael, Nando de Freitas, and Shimon Whiteson. Learning to communicate with deep multi-agent reinforcement learning. In *NeurIPS*, 2016.
[26] Sainbayar Sukhbaatar, Arthur Szlam, and Rob Fergus. Learning multiagent communication with backpropagation. In *NeurIPS*, 2016.
[27] Brendon Boldt and David Mortensen. A review of the applications of deep learning-based emergent communication. *TMLR*, 2024.
[28] Noga Zaslavsky, Jennifer Hu, and Roger P. Levy. A rate-distortion view of human pragmatic reasoning. In *SCiL*, 2021.
[29] Jacob Andreas and Dan Klein. Reasoning about pragmatics with neural listeners and speakers. In *EMNLP*, 2016.
[30] Daniel Fried, Jacob Andreas, and Dan Klein. Unified pragmatic models for generating and following instructions. In *NAACL-HLT*, 2018.
[31] Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, and Sergey Levine. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. In *ICML*, 2018.
[32] Sergey Levine. Reinforcement learning and control as probabilistic inference: Tutorial and review. *arXiv:1805.00909*, 2018.
[33] David Barber and Felix Agakov. The IM algorithm: A variational approach to information maximization. In *NeurIPS*, 2003.
[34] Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. Multi-agent actor-critic for mixed cooperative-competitive environments. In *NeurIPS*, 2017.
[35] Samuel Sokota, Christian A. Schroeder de Witt, Maximilian Igl, Luisa M. Zintgraf, Philip Torr, Martin Strohmeier, Zico Kolter, Shimon Whiteson, and Jakob Foerster. Communicating via Markov decision processes. In *ICML*, 2022.
[36] Mark K. Ho, Michael Littman, James MacGlashan, Fiery Cushman, and Joseph L. Austerweil. Showing versus doing: Teaching by demonstration. In *NeurIPS*, 2016.
[37] Jaime F. Fisac, Monica A. Gates, Jessica B. Hamrick, Chang Liu, Dylan Hadfield-Menell, Malayandi Palaniappan, Dhruv Malik, S. Shankar Sastry, Thomas L. Griffiths, and Anca D. Dragan. Pragmatic-pedagogic value alignment. In *ISRR*, 2017.
[38] Baptiste Busch, Jonathan Grizou, Manuel Lopes, and Freek Stulp. Learning legible motion from human–robot interactions. *Int. J. Social Robotics*, 9(5):765–779, 2017.
[39] Manuel Bied and Mohamed Chetouani. Integrating an observer in interactive reinforcement learning to learn legible trajectories. In *RO-MAN*, 2020.
[40] Hugo Caselles-Dupré, Olivier Sigaud, and Mohamed Chetouani. Pragmatically learning from pedagogical demonstrations in multi-goal environments. In *NeurIPS*, 2022.
[41] Michele Persiani and Thomas Hellström. Policy regularization for legible behavior. *Neural Computing and Applications*, 35:16781–16790, 2023.
[42] Sebastian Wallkötter, Mohamed Chetouani, and Ginevra Castellano. SLOT-V: Supervised learning of observer models for legible robot motion planning. In *RO-MAN*, 2022.
[43] Woojun Kim, Jongeui Park, and Youngchul Sung. Communication in multi-agent reinforcement learning: Intention sharing. In *ICLR*, 2021.
[44] Reuth Mirsky, Ignacio Carlucho, Arrasy Rahman, Elliot Fosong, William Macke, Mohan Sridharan, Peter Stone, and Stefano V. Albrecht. A survey of ad hoc teamwork research. In *EUMAS*, 2022.
[45] Zheng Tian, Shihao Zou, Ian Davies, Tim Warr, Lisheng Wu, Haitham Bou Ammar, and Jun Wang. Learning to communicate implicitly by actions. In *AAAI*, 2020.
[46] Yipeng Kang, Tonghan Wang, and Gerard de Melo. Incorporating pragmatic reasoning communication into emergent language. In *NeurIPS*, 2020.
[47] Ross A. Knepper, Christoforos I. Mavrogiannis, Julia Proft, and Claire Liang. Implicit communication in a joint action. In *HRI*, 2017.
[48] Will Monroe and Christopher Potts. Learning in the rational speech acts model. In *Amsterdam Colloquium*, 2015.
[49] Ryan Lowe, Jakob Foerster, Y-Lan Boureau, Joelle Pineau, and Yann Dauphin. On the pitfalls of measuring emergent communication. In *AAMAS*, 2019.
[50] Han Wang, Binbin Chen, Tieying Zhang, and Baoxiang Wang. Learning to communicate through implicit communication channels. In *ICLR*, 2025.
[51] Miquel Ramírez and Hector Geffner. Probabilistic plan recognition using off-the-shelf classical planners. In *AAAI*, 2010.
[52] Sarah Keren, Avigdor Gal, and Erez Karpas. Goal recognition design. In *ICAPS*, 2014.
[53] Dylan Hadfield-Menell, Anca D. Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *NeurIPS*, 2016.
[54] Hong Jun Jeon, Smitha Milli, and Anca D. Dragan. Reward-rational (implicit) choice: A unifying formalism for reward learning. In *NeurIPS*, 2020.
[55] Smitha Milli and Anca D. Dragan. Literal or pedagogic human? Analyzing human model misspecification in objective learning. In *UAI*, 2019.
[56] Andrew Fuchs, Michael Walton, Theresa Chadwick, and Doug Lange. Theory of mind for deep reinforcement learning in Hanabi. *arXiv:2101.09328*, 2021.
[57] Ini Oguntola, Joseph Campbell, Simon Stepputtis, and Katia Sycara. Theory of mind as intrinsic motivation for multi-agent reinforcement learning. In *ICML Workshop on Theory of Mind in Communicating Agents*, 2023.